

← History

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❑ clc;
clear;
close all;
% System Parameters
Bandwidth = 10e6; % Total bandwidth
(10 MHz)
RB_size = 180e3; % Resource Block
size (180 kHz)
Total_RBs = floor(Bandwidth/RB_size);
Users = 10; % Number of active
users
% Equal RB Allocation
RB_alloc = floor(Total_RBs/Users) *
ones(1,Users);
% Allocate remaining RBs
remaining = Total_RBs - sum(RB_alloc);
for i = 1:remaining
    RB_alloc(i) = RB_alloc(i) + 1;
end
% Resource Block Utilization
Allocated_RBs = sum(RB_alloc);
RB_utilization = (Allocated_RBs/
Total_RBs)*100;
% Modulation Efficiency (bits/s/Hz)
% QPSK = 2, 16-QAM = 4, 64-QAM = 6
Efficiency = [2 4 6 2 4 6 2 4 6 4];
% Throughput Calculation
RB_bandwidth = 180e3;
Throughput = (RB_alloc .* RB_bandwidth .*
Efficiency)/1e6;
% Display Results
fprintf('Total Bandwidth = %.0f
MHz\n',Bandwidth/1e6);
fprintf('Total Resource Blocks =
%d\n',Total_RBs);
fprintf('Allocated Resource Blocks =
%d\n',Allocated_RBs);
fprintf('Resource Block Utilization = %.2f %%
\n\n',RB_utilization);
disp('User    RBs Allocated    Spectral
Efficiency    Throughput (Mbps)')
for i = 1:Users

fprintf('%2d\t\t%2d\t\t\t%2d\t\t\t%.2f\n', ...

i,RB_alloc(i),Efficiency(i),Throughput(i));
end
% Plot 1: RB Allocation
figure;
bar(1:Users,RB_alloc);
xlabel('Users');
ylabel('Allocated Resource Blocks');
title('OFDMA Resource Block Allocation');
grid on;
% Plot 2: User Throughput

```

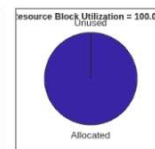
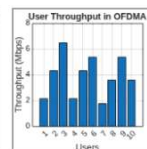
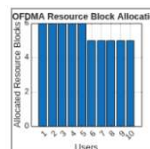
11:07

VoLTE 4G+ LTE



Total Bandwidth = 10 MHz
 Total Resource Blocks = 55
 Allocated Resource Blocks = 55
 Resource Block Utilization = 100.00 %

User	RBs	Allocated	Spectral Efficiency
Throughput (Mbps)			
1	6	2	2.16
2	6	4	4.32
3	6	6	6.48
4	6	2	2.16
5	6	4	4.32
6	5	6	5.40
7	5	2	1.80
8	5	4	3.60
9	5	6	5.40
10	5	4	3.60



>> Enter command





Figure 1: clc; clear; close all; % System Parameters Bandwidth = 1...

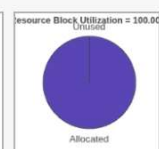
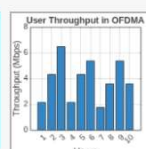
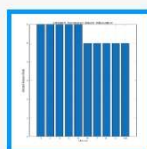
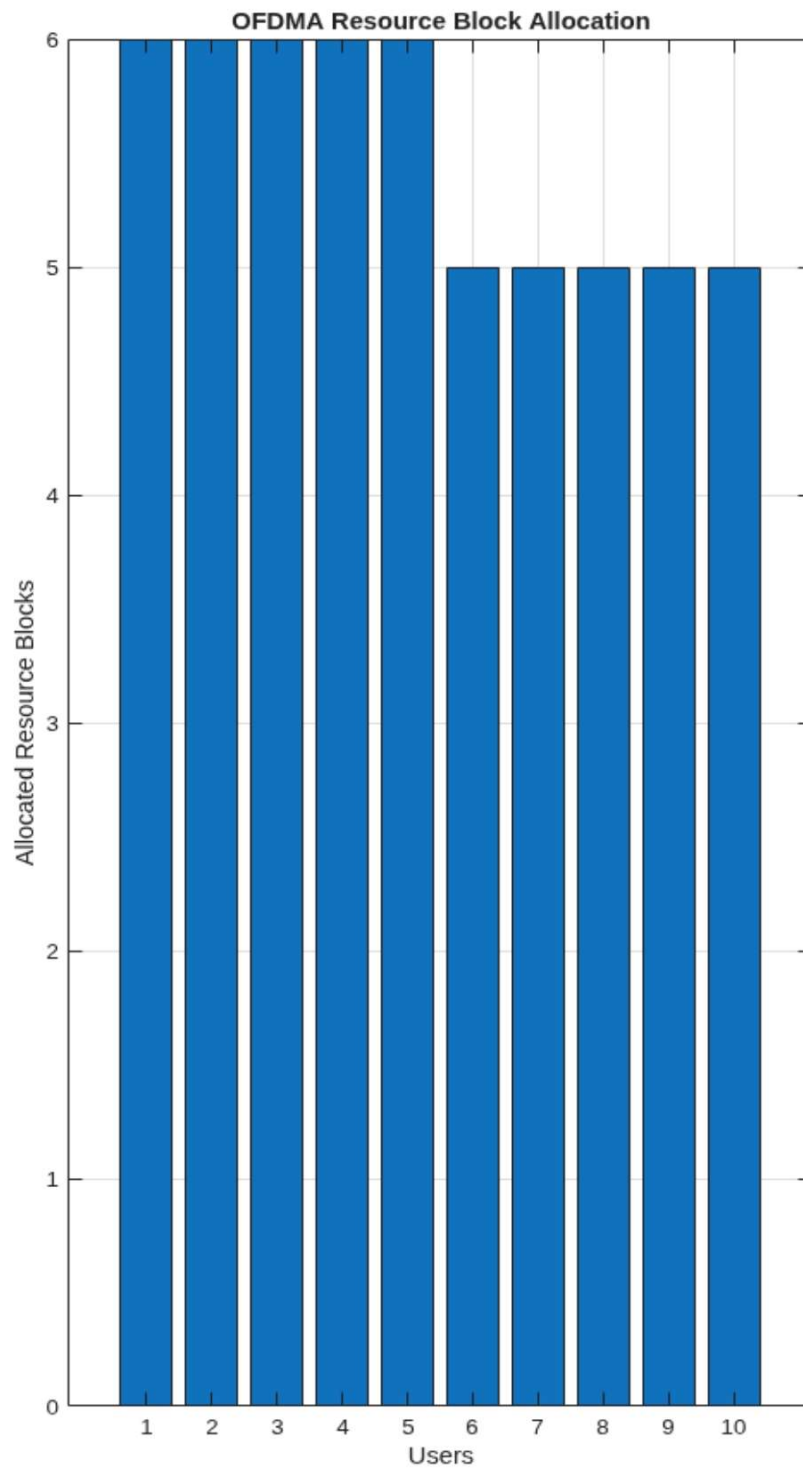
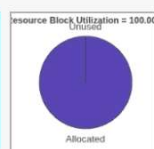
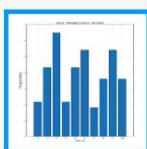
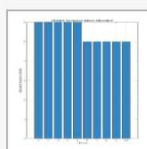
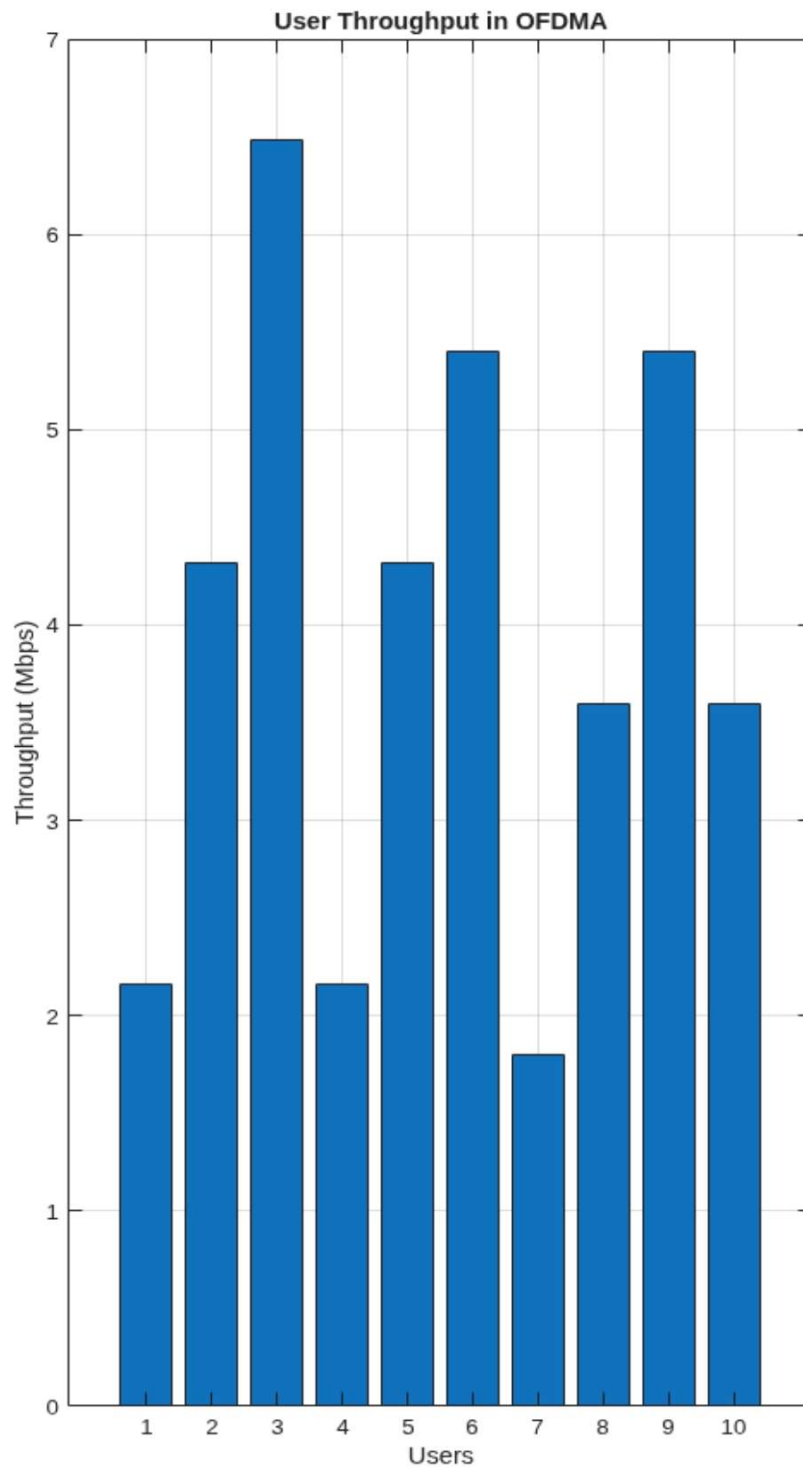




Figure 2: clc; clear; close all; % System Parameters Bandwidth = 1...



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Vol 4G+
LTE



Figures

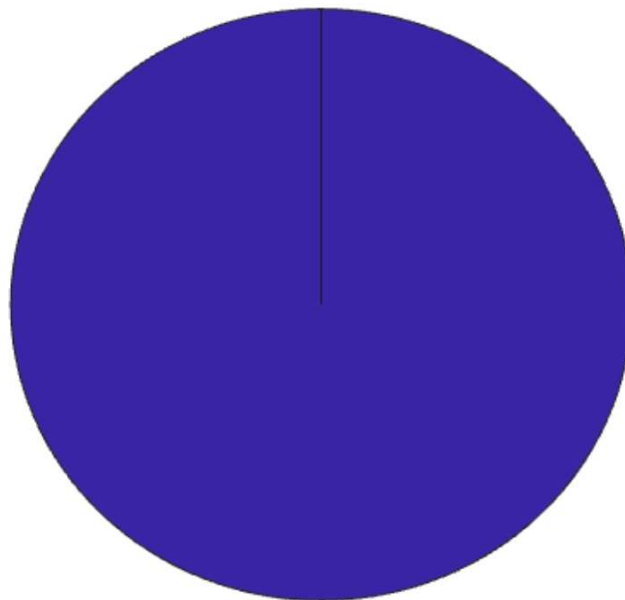
SHOW DATA CURSOR



Figure 3: clc; clear; close all; % System Parameters Bandwidth = 1...

Resource Block Utilization = 100.00%

Unused



Allocated

